

Optimized Classification Model - E-Commerce

Purchase Prediction Report

1. Executive Summary

This report presents the results of an optimized classification model built to predict customer purchase likelihood for an e-commerce platform. The XGBoost model achieved an accuracy of 97.60% and an F1-Score of 0.9800, demonstrating strong predictive power. The model identifies CartItems and TimeOnSite as the most influential features. Business recommendations are provided to target customers based on their purchase risk category.

2. Dataset Overview

Total Records : 10000
Total Features : 18
Purchase Rate : 59.68%
Target Variable : Purchase (Binary: 0 = Not Bought, 1 = Bought)
Key Features : CartItems, TimeOnSite, PagesViewed, EmailClicked
Class Balance : Balanced

3. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.9760	0.9743	0.9858	0.9800	0.9763
Decision Tree	0.9715	0.9749	0.9774	0.9762	0.9629
Random Forest	0.9795	0.9800	0.9858	0.9829	0.9788
XGBoost	0.9765	0.9759	0.9849	0.9804	0.9794
XGBoost (Optimized)	0.9765 Threshold)	0.9751	0.9858	0.9804	0.9793

4. Best Model (XGBoost) Metrics

Optimization Status : Optimized
Accuracy : 0.9760 (97.60%)

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Precision : 0.9743
Recall : 0.9858
F1-Score : 0.9800
ROC-AUC : 0.9793
Optimal Threshold : 0.60

5. Top 5 Influential Features

5. num__CartItems : 0.7863

20. cat__DiscountUsed_1 : 0.0115

17. cat__TrafficSource_Email : 0.0115

19. cat__TrafficSource_Social : 0.0111

14. cat__DeviceType_Mobile : 0.0111

6. Business Recommendations

High Likelihood Customers (Risk: High) :

- Strategy: Aggressive Conversion
- Actions: 1-Click checkout, Cart reminders, Exclusive offers

Medium Likelihood Customers (Risk: Medium) :

- Strategy: Nurturing Campaign
- Actions: Product recommendations, Reviews, Free shipping incentive

Low Likelihood Customers (Risk: Low) :

- Strategy: Brand Awareness
- Actions: Low-cost emails, Collect feedback, Improve UX

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7. Conclusion

The optimized XGBoost model provides a reliable method for predicting customer purchases.

Feature importance analysis reveals that CartItems and TimeOnSite are the primary drivers of purchase decisions.

Implementing the recommended business strategies can significantly improve conversion rates and marketing ROI.

The model is production-ready and can be deployed via the saved pickle pipeline.